

# Beginner Basics of C#

## Module 5 - C# IF, Switch, For, While loop Statement

Both **For** and **While** loops allow you to repeat a block of code multiple times, but they are used in different scenarios based on whether you know exactly how many times the code needs to run.

### The While Loop

A while loop is used when the number of iterations is **unknown** beforehand. It keeps running as long as a specific condition remains true. You use it when you say, "Keep running until the user types 'exit'" or "Keep running until the player's health reaches zero."

**Structure:** It only contains the condition. You must manage the changes to that condition inside the loop body.

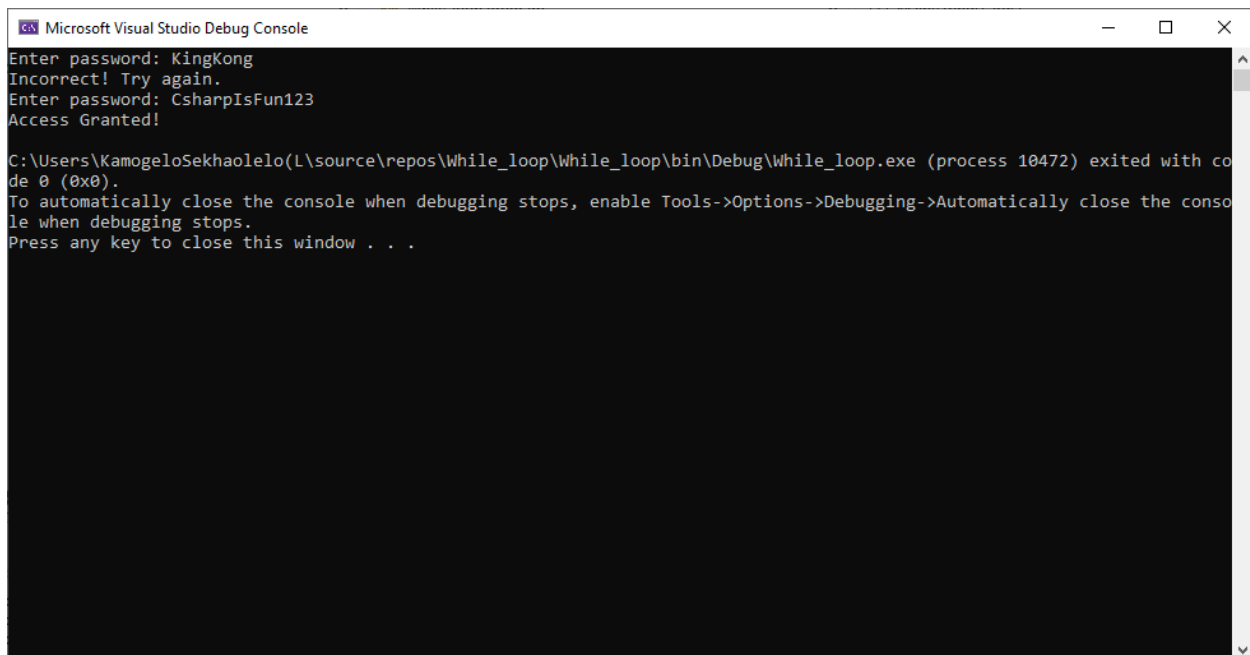
### Example: The while Loop (User Interaction)

Now imagine a login screen. You want to keep asking the user for a password until they get it right. You have no idea if they will get it on the 1st try or the 100th try, so you use a while loop.



```
27
28
29 string correctPassword = "CsharpIsFun123";
30 string userAttempt = "";
31
32 // Keep looping as long as the attempt does NOT match the correct password
33 while (userAttempt != correctPassword)
34 {
35     Console.Write("Enter password: ");
36     userAttempt = Console.ReadLine();
37
38     if (userAttempt != correctPassword)
39     {
40         Console.WriteLine("Incorrect! Try again.");
41     }
42
43     Console.WriteLine("Access Granted!");
44 }
45
```

**OutPut:**

A screenshot of the Microsoft Visual Studio Debug Console window. The window has a title bar with the text "Microsoft Visual Studio Debug Console" and standard window controls (minimize, maximize, close). The console output is as follows:  
Enter password: KingKong  
Incorrect! Try again.  
Enter password: CsharpIsFun123  
Access Granted!  
  
C:\Users\KamogeloSekhaolelo(L\source\repos\While\_loop\While\_loop\bin\Debug\While\_loop.exe (process 10472) exited with code 0 (0x0).  
To automatically close the console when debugging stops, enable Tools->Options->Debugging->Automatically close the console when debugging stops.  
Press any key to close this window . . .

As you can see from the example above, I displayed 2 inputs which is both correct and incorrect password. Every time I enter an incorrect password the "Try again." Prompt will continue to loop indefinitely until I enter a correct password then the loop stops

### The For Loop:

A for loop is generally used when you have a **fixed number of iterations**. You use it when you can say, "Run this code exactly 10 times" or "Run this for every item in this list."

#### Example: The for Loop (Processing Data)

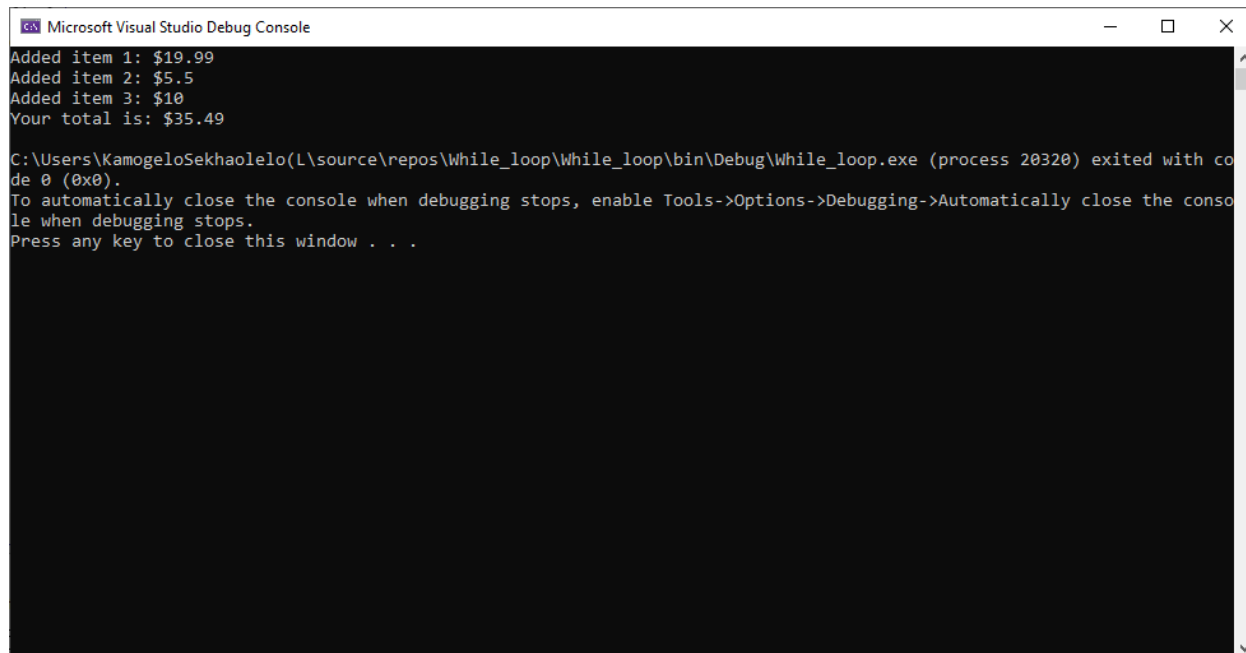
Imagine you are building a simple shopping cart. You have an array (a list) of item prices, and you need to calculate the total. Since you know exactly how many items are in the list, a for loop is perfect.

```

45
46
47
48     double[] prices = { 19.99, 5.50, 10.00 };
49     double total = 0;
50
51     // We know there are exactly 3 items, so we loop 3 times
52     for (int i = 0; i < prices.Length; i++)
53     {
54         total += prices[i];
55         Console.WriteLine("Added item " + (i + 1) + ": $" + prices[i]);
56     }
57
58     Console.WriteLine("Your total is: $" + total);
59
60     }
61 }

```

## OutPut:



The screenshot shows the Microsoft Visual Studio Debug Console window. The output text is as follows:

```

Added item 1: $19.99
Added item 2: $5.5
Added item 3: $10
Your total is: $35.49

C:\Users\KamogeloSekhaolelo(L\source\repos\While_loop\While_loop\bin\Debug\While_loop.exe (process 20320) exited with code 0 (0x0).
To automatically close the console when debugging stops, enable Tools->Options->Debugging->Automatically close the console when debugging stops.
Press any key to close this window . . .

```